Multiple IV Bolus Practice 1

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a One gram of a drug was administered by IV bolus injection every 8 hours to a 65-kg patient. The drug has an apparent V_D of 26.2 L/100 kg. The total body clearance is 52.95 mL/min per 70 kg and renal clearance is 34.72 mL/min per 70 kg. Calculate the elimination half-life.

1b One gram of a drug was administered by IV bolus injection every 8 hours to a 65-kg patient. The drug has an apparent V_D of 26.2 L/100 kg. Calculate the maximum plasma drug concentration of the first dose.

1c One gram of a drug was given by IV bolus every 8 hours. The first dose peaks at 58.72 mg/L and k is 0.1732 hr^{-1}. Calculate the minimum plasma drug concentration of the first dose.

1d One gram of a drug is given by IV bolus every 8 hours. The first dose peaks at 58.72 mg/L and k is 0.1732 hr^{-1}. Calculate the maximum steady-state plasma drug concentration.

1e A drug given by IV bolus every 8 hours reaches 78.31 mg/L at its steady-state peak, and k is 0.1732 hr^{-1}. Calculate the minimum steady-state plasma drug concentration.

1f One gram of a drug is given by IV bolus every 8 hours to a patient whose total body clearance is 2.95 L/hr. Calculate the average steady-state plasma drug concentration.

1g A drug given by repeated IV bolus has an average steady-state concentration of 42.37 mg/L and an apparent V_D of 17.03 L. Calculate the average amount of drug in the body during steady state.

1h A drug given by IV bolus every 8 hours has a steady-state peak of 78.31 mg/L and k of 0.1732 hr^{-1}. Calculate the plasma level 12 hours after the last dose, assuming steady-state levels are achieved.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a

Her key

Answer: 4 hr

WHAT THE STEM GIVES

D_0	One gram	*not needed: the half-life does not depend on the dose*
τ	every 8 hours	*not needed: half-life is a drug constant*
weight	65-kg	scales both Cl_T and V_D to this patient
V_D	26.2 L/100 kg	scaled to 65 kg: 17.03 L, the denominator of k
Cl_T	52.95 mL/min per 70 kg	scaled and converted: 2.95 L/hr, the numerator of k
Cl_R	34.72 mL/min per 70 kg	*not needed: elimination half-life uses total clearance*

THE WORKING, ONE LINE AT A TIME

UNIT

$$Cl_T = \frac{52.95\ \text{mL/min}}{70\ \text{kg}} \times 65\ \text{kg} \times \frac{1\ \text{L}}{1000\ \text{mL}} \times \frac{60\ \text{min}}{1\ \text{hr}} = 2.95\ \text{L/hr}$$

The clearance is given per 70 kg, so it is scaled to this 65-kg patient first. mL to L and minutes to hours put it in the same units as the volume and the answer.

UNIT

$$V_D = \frac{26.2\ \text{L}}{100\ \text{kg}} \times 65\ \text{kg} = 17.03\ \text{L}$$

The volume is also given per unit weight, per 100 kg here, so it is scaled to 65 kg the same way: 26.2 × 0.65 = 17.03 L.

SETUP

$$k = \frac{Cl_T}{V_D};\ t½ = \frac{0.693}{k}$$

Clearance is k times V_D, so k is clearance divided by volume, and the half-life follows from k. Neither step needs the dose or the interval; they come in only from part (b) on.

ALGEBRA

$$k = \frac{2.95\ \text{L/hr}}{17.03\ \text{L}} = 0.1732\ \text{hr}^{-1};\ t½ = \frac{0.693}{0.1732\ \text{hr}^{-1}} = 4\ \text{hr}$$

In $k = \frac{Cl_T}{V_D}$ the litres cancel, leaving reciprocal hours; 0.693 divided by reciprocal hours leaves hours. $\frac{0.693}{0.1732} = 4.00$.

ROUND

$$t½ = 4\ \text{hr}$$

Her keyed value, 4 hr. With τ = 8 hr the interval is two half-lives, the same as in Example 1, so the accumulation factor in the later parts is again 1.33.

DOES THE NUMBER MAKE SENSE?

$k = \frac{2.95}{17.03} = 0.1732\ hr^{-1}$, so t½ has to come out near $\frac{0.693}{0.1732} = 4$ hr. The per-70-kg clearance and the per-100-kg volume both scale by the same 65 kg, so the ratio barely moves.

1b Her key

Answer: 58.72 mg/L

WHAT THE STEM GIVES

D_0	One gram	1000 mg, the numerator
τ	every 8 hours	*not needed: no drug precedes the first dose*
weight	65-kg	scales the per-100-kg volume to 17.03 L
V_D	26.2 L/100 kg	scaled to 65 kg: 17.03 L, the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = 1\ g = 1000\ mg;\ V_D = \frac{26.2\ L}{100\ kg} \times 65\ kg = 17.03\ L$$

Grams to milligrams so the answer is in mg/L, and the volume scaled to the patient, because it is given per 100 kg of body weight.

SETUP

$$C_{max}\ (\text{first dose}) = \frac{D_0}{V_D}$$

The first bolus enters a body with no drug, so its peak is simply C_0, the whole dose divided by the volume at time zero.

ALGEBRA

$$C_{max} = \frac{1000\ mg}{17.03\ L} = 58.72\ mg/L$$

1000 divided by 17.03 is 58.72, and mg over L is a concentration. This C_0 is the starting value for every later part.

ROUND

$$C_{max} = 58.72\ mg/L$$

Her keyed value, 58.72 mg/L, to two decimal places, the precision she carries through the rest of the set. It is also the first-dose C_{max}.

DOES THE NUMBER MAKE SENSE?

A gram spread through about 17 L is a little under 60 mg/L; $\frac{1000\ mg}{17.03\ L} = 58.72$ mg/L. The dose is above zero, so the peak is too.

1c Her key

Answer: 14.69 mg/L

WHAT THE STEM GIVES

D_0	One gram	*not needed: already inside the given peak*
τ	every 8 hours	the time in $e^{-k\tau}$, with $k\tau = 0.1732 \times 8 = 1.386$
C_0	58.72 mg/L	the first-dose peak the trough is read down from
k	0.1732 hr^{-1}	the exponent $k\tau$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min} = C_{max}\, e^{-k\tau}$$

After the first dose the level falls by first-order elimination until the next dose, so the minimum is at $t = \tau = 8$ hr.

ALGEBRA

$$C_{min} = 58.72 \text{ mg/L} \times e^{-(8 \text{ hr})(0.1732 \text{ hr}^{-1})} = 58.72 \times 0.2502$$

$k\tau = 8 \times 0.1732 = 1.386$, a pure number. The interval is two half-lives, so about a quarter of the peak remains.

ALGEBRA

$$C_{min} = 14.69 \text{ mg/L}$$

$58.72 \times 0.2502 = 14.69$. The same result comes from halving twice: 58.72, 29.36, 14.68, with the last digit set by rounding.

ROUND

$$C_{min} = 14.69 \text{ mg/L}$$

Her keyed value, 14.69 mg/L. It is the first-dose trough, and the steady-state trough in part (e) is higher by the accumulation factor.

> **DOES THE NUMBER MAKE SENSE?**
>
> $k = 0.1732$ hr^{-1} means a half-life of $\frac{0.693}{0.1732} = 4$ hr, so 8 hours is two half-lives and a quarter of the peak remains: 58.72×0.25 is just under 14.69 mg/L, below the 58.72 mg/L peak.

1d

Her key

Answer: 78.31 mg/L

WHAT THE STEM GIVES

D_0	One gram	*not needed: already inside the given peak*
τ	every 8 hours	the exponent $k\tau = 0.1732 \times 8 = 1.386$
C_0	58.72 mg/L	the numerator
k	0.1732 hr^{-1}	the exponent $k\tau$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}}$$

The first-dose peak divided by $(1 - e^{-k\tau})$ adds the drug accumulated from earlier doses, so the result must be larger than 58.72 mg/L.

ALGEBRA

$$e^{-(8\ hr)(0.1732\ hr^{-1})} = e^{-1.386} = 0.2502;\ 1 - 0.2502 = 0.7498$$

$e^{-k\tau}$ is the fraction of each dose left at the end of an interval, and 1 minus it is the fraction removed during the interval.

ALGEBRA

$$C_{max}^{\infty} = \frac{58.72\ mg/L}{1 - e^{-(8)(0.1732)}} = \frac{58.72\ mg/L}{0.7498} = 78.31\ mg/L$$

Dividing by a number less than 1 raises the peak, as accumulation must: $\frac{58.72}{0.7498} = 78.31$, a factor of 1.33.

ROUND

$$C_{max}^{\infty} = 78.31\ mg/L$$

Her keyed value, 78.31 mg/L, higher than the first-dose 58.72 mg/L by the accumulation factor 1.33, as a steady-state peak must be.

DOES THE NUMBER MAKE SENSE?

Eight hours is two half-lives at $k = 0.1732\ hr^{-1}$, so a quarter of each dose remains at the next; the accumulation factor is $\frac{1}{1 - 0.25}$, a third more than C_0: 78.31 mg/L, above 58.72 and below 2×58.72.

1e

Her key

Answer: 19.59 mg/L

WHAT THE STEM GIVES

τ	every 8 hours	the time in $e^{-k\tau}$
C_{max}^{∞}	78.31 mg/L	the steady-state peak the trough is read down from
k	$0.1732\ hr^{-1}$	the exponent $k\tau = 0.1732 \times 8 = 1.386$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = C_{max}^{\infty} e^{-k\tau}$$

From the steady-state peak to the steady-state trough is one interval of first-order decline, so the peak is multiplied by $e^{-k\tau}$.

ALGEBRA

$$C_{min}^{\infty} = 78.31\ mg/L \times e^{-(8)(0.1732)} = 78.31 \times 0.2502$$

The same fraction remains as after the first dose, 0.2502, because k and τ have not changed; only the starting level is higher.

ALGEBRA

$C_{min}^{\infty} = 19.59$ mg/L

78.31 × 0.2502 = 19.59. Checked the other way, 14.69 × 1.3336 = 19.59, the first-dose trough times the accumulation factor.

ROUND

$C_{min}^{\infty} = 19.59$ mg/L

Her keyed value, 19.59 mg/L. It is above the first-dose trough of 14.69 mg/L, which is the drug that has accumulated.

DOES THE NUMBER MAKE SENSE?

Eight hours is two half-lives at k = 0.1732 hr^{-1}, so a quarter of the peak remains: 78.31 × 0.25 is just under 19.59 mg/L. A trough lies below its peak.

1f Her key

Answer: 42.37 mg/L

WHAT THE STEM GIVES

D_0	One gram	1000 mg, the numerator; F = 1 for IV
τ	every 8 hours	multiplies clearance: Cl_T τ = 2.95 × 8 = 23.6 L
Cl_T	2.95 L/hr	the denominator with τ

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}, \text{ with } F = 1$$

With clearance already known from part (a), the Cl_T form is the direct one; it is the same as $\frac{FD_0}{V_D\,k\tau}$ because $Cl_T = kV_D$. intravenous (IV) dosing gives F = 1.

ALGEBRA

$$C_{avg}^{\infty} = \frac{1000\text{ mg}}{(2.95\text{ L/hr})(8\text{ hr})}$$

In the denominator L/hr times hr leaves L, so mg over L gives a concentration in mg/L. The dose is 1000 mg, one gram converted to milligrams.

ALGEBRA

$$C_{avg}^{\infty} = \frac{1000\text{ mg}}{23.6\text{ L}} = 42.37\text{ mg/L}$$

2.95 × 8 = 23.6 L, the volume of plasma cleared of drug in one interval, and 1000 mg divided by 23.6 L is 42.37 mg/L.

ROUND

$C_{avg}^{\infty} = 42.37$ mg/L

Her keyed value, 42.37 mg/L. It lies between the trough, 19.59, and the peak, 78.31, and below their midpoint, 48.95, because the decline is exponential.

DOES THE NUMBER MAKE SENSE?

Clearance of 2.95 L/hr for 8 hours clears 23.6 L, so the gram is spread through 23.6 L: $\frac{1000\ \text{mg}}{23.6\ \text{L}} = 42.37\ \text{mg/L}$. The dose is above zero, so the average is too.

1g Her key

Answer: 721.6 mg

WHAT THE STEM GIVES

C_{avg}^{∞}	42.37 mg/L	multiplied by V_D
V_D	17.03 L	multiplied by C_{avg}^{∞}

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_{avg}^{\infty} = C_{avg}^{\infty} \times V_D$$

Amount and concentration are linked by the volume of distribution at every time, including on average at steady state.

ALGEBRA

$$D_{avg}^{\infty} = (42.37\ \text{mg/L})(17.03\ \text{L})$$

mg/L times L: the litres cancel, leaving mg, an amount, which is what the question asks for. The volume is the 17.03 L found in part (a).

ALGEBRA

$$D_{avg}^{\infty} = 721.6\ \text{mg}$$

$42.37 \times 17.03 = 721.6$. The same result comes from the amount form, $\frac{FD_0}{k\tau} = \frac{1000}{0.1732 \times 8} = 721.7$, within rounding.

ROUND

$$D_{avg}^{\infty} = 721.61\ \text{mg}$$

Her keyed value is 721.61 mg, and 721.6 mg is the same to four significant figures. It lies between the minimum and maximum amounts at steady state, $D_{min}^{\infty} = 1000 \times \frac{0.2502}{0.7498} = 333.7$ mg and $D_{max}^{\infty} = \frac{1000}{0.7498} = 1333.7$ mg.

DOES THE NUMBER MAKE SENSE?

A concentration times a volume is an amount: 42.37 mg/L × 17.03 L = 721.6 mg, in milligrams because the litres cancel. The amount is above zero.

1h Her key

Answer: 9.799 mg/L

WHAT THE STEM GIVES

τ	every 8 hours	*not needed: the time asked is counted from the last dose*
C_{max}^{∞}	78.31 mg/L	the peak the level decays from

k	0.1732 hr^{-1}	the exponent kt = 0.1732 × 12 = 2.078
t	12 hours after the last dose	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_{max}{}^{\infty}e^{-kt}$, with t = 12 hr

After the last dose at steady state the level falls from the steady-state peak by first-order elimination. The 12 hours is measured from the last dose.

ALGEBRA

$e^{-(12\ hr)(0.1732\ hr^{-1})} = e^{-2.078} = 0.1251$

12 × 0.1732 = 2.078, a pure number. Twelve hours is three half-lives, so one eighth remains: $e^{-2.078}$ = 0.1251.

ALGEBRA

C = 78.31 mg/L × 0.1251 = 9.799 mg/L

78.31 × $e^{-2.078}$ = 9.799. Dividing 78.31 by 8 gives 9.79, the same result by halving three times: 39.16, 19.58, 9.79.

ROUND

C = 9.799 mg/L

Her keyed value, 9.799 mg/L. It is below the steady-state trough of 19.59 mg/L because 12 hours is longer than the 8-hour interval.

> **DOES THE NUMBER MAKE SENSE?**
> k = 0.1732 hr^{-1} is a 4-hour half-life, so 12 hours is three half-lives and the peak halves three times: 78.31 × 0.125 is just under 9.799 mg/L, an eighth of the peak and well below it.